Zhangjin Zhao

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Research / R&D Internship | Available from Dec. 2026

Homepage & CV

EDUCATION

Southern University of Science and Technology (SUSTech) U.S. News Global 2025–26: #123
Master's student, School of Microelectronics Sep. 2025–Jun. 2028 (expected)

- **Advisor:** Prof. Quan Chen
- **Research focus:** AI for EDA
  - AI agents for netlist exploration
  - Multi-conductor S-parameter prediction for PCB signal integrity
  - Analog circuit sizing automation
- **Highlights:** Honorable Mention Prize, ICCAD 2026 CAD Contest; 3 manuscripts under review (2 DAC, 1 DATE).

Bachelor's, Microelectronics Science and Engineering, SUSTech Sep. 2021–Jun. 2025

RESEARCH AND PROJECT EXPERIENCE

Verifiable LLM Agent Framework for EDA (Netlist Case Study)
Team lead | ***ICCAD 2026 CAD Contest (Cadence), Honorable Mention Prize*** | *First-author manuscript under review at DAC*

- Built a verifiable, tool-augmented LLM agent for EDA: from a gate-level netlist and sequential natural-language requests, it performs analysis, transformation and constrained optimization with checkable results.
- Neuro-symbolic: LLM semantic parser and planner emitting schema-typed tool calls (81 composable operators) over a persistent state; edits follow propose–verify–commit with Yosys/ABC equivalence checks and rollback.
- Scored 661.4/674 (98.1%) on 100 hidden test cases (netlists up to 213k instances; official best-of-two-LLM scoring). Operators, state and verifiers are pluggable, so the loop extends to other EDA tasks.

Multi-Conductor S-Parameter Prediction for PCB Signal Integrity
Co-first-author manuscript under review at DAC | Prof. Quan Chen's group, SUSTech | Collaborations: Xpeedic, OPPO

- Develop graph-Transformer surrogates that predict complex multiport S-parameters of 4–16-conductor PCB interconnects from layout geometry, stack-up/materials and frequency, replacing repeated full-wave EM extraction.
- Use a GRIT graph Transformer with propagation-delay encodings, frequency-conditioned cross-attention and port-pair readouts; the 4-conductor model predicts 36 port pairs at 100 frequencies (0.5–50 GHz).
- Inference takes milliseconds versus hours for commercial EM extraction (DDR4 byte, three PCIe 5 differential pairs); 0.128 dB magnitude MAE and 0.874° circular phase MAE vs PowerSI on 358 held-out layouts.

Data-Efficient Cross-Technology Analog IC Sizing
First author, manuscript under review at DATE 2027 | ASTRI EDA Elite Challenge project

- Developed a cross-technology inverse sizing framework for analog ICs: given performance specifications, it generates op-amp sizing candidates for a new process node from scarce target-process data.
- Pretrained a specification-conditioned Diffusion Transformer on source-process designs, adapted it with LoRA, and screened candidates with a transferred multi-output MLP before SPICE verification (generate–screen–verify); built the OCEAN/Spectre datasets for 3 op-amp topologies at 180/130/55 nm.
- Needed 19.8–71.3% fewer target samples than full fine-tuning at <2% mean relative error (10 metrics, 5 transfers, 3 seeds); top-20-of-100 screening cut SPICE evaluations by 80%.

SELECTED AWARDS

Honorable Mention Prize – 2026 CAD Contest at ICCAD 2026
Problem A: LLM-Assisted Netlist Exploration and Transformation (team cada1078)
About 7% of registered teams win awards (2021–23 contest statistics).

National Second Prize – China Postgraduate IC Innovation Competition (674 teams overall) 2025
EDA Elite Challenge, Problem 9: AI-Based Cross-Technology Behavioral Modeling of Analog ICs (setter: ASTRI).

University-Level Outstanding Graduate Student, SUSTech 2026
Graduate Academic Scholarship & Research Assistantship Stipend, both Highest Tier 2026–27

TECHNICAL SKILLS

ML & Agents: OpenAI/Anthropic tool calling, JSON-schema tool design, stateful agents, LLM evaluation; PyTorch, PyTorch Geometric, graph Transformers, DiT/LoRA.
Languages: Python, Verilog, Tcl (PowerSI batch scripting), OCEAN scripting.
EDA & SI: Cadence PowerSI/Spectre, Ansys HFSS, Yosys/ABC equivalence checking, Icarus Verilog, Touchstone.
Development: Linux, Git; GPU model training/evaluation, mixed-precision PyTorch.